

Malware Similarity Trends and Gaps Analysis

Progress Report Presentation



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Project Summary

Overview

- The main goal of this project is to develop an **automated tool that analyzes malware** and outputs useful information about the malware.
- We have researched the **malware development life cycle (MDLC)** and each written our own malware in various languages based on provided requirements for an **exfiltration tool**.
- We will compare the **final product** of our malware tools, in the form of Windows executables, to find **similarities and differences evident from each phase** of the MDLC.
- Based on our analysis, we will have a basis to begin developing our prototype to **extract these features automatically** from a piece of malware.

Work Completed So Far

- **Exfiltration Tools:** The team has developed variations of an exfiltration tool based on provided requirements that accomplishes the following tasks:
 - Extracts files specified by the attacker from a victim's machine.
 - Obfuscates these files.
 - Sends the obfuscated files to an IP address or FQDN through a HTTP Post command.
- **Deobfuscation Tools:** The tool comes with an additional deobfuscation tool or option so that once the adversary retrieves these files from the IP address or FQDN, the exfiltrated files are deobfuscated.
- **Malware Implementation Variations:** The team has developed six exfiltration tools, in which each team member developed two implementations in two different languages.
 - Three variations in Python.
 - Two variations in Java.
 - One variation in Go.
- **Malware Analysis Environment:** The team has setup remote access to two computers.
 - One computer has a REMnux Virtual Machine installed that will assist the team with malware analysis.

Work In Progress

- **Packaging and Obfuscation:** We are working on finding obfuscation techniques to obfuscate the developed malware using various packaging methods and tools.
 - Specifically, we are working to package all of the exfiltration tools as standalone Windows executables with various configuration options such that an attacker can directly run the executable on the victim's machine.
 - This specific goal is important because malware analysts typically only work with the final product or self-contained executable version of the malware rather than analyzing the malware at each phase of its development.
- **Determination of Features Detectable in Executables:** Once this task is completed, we will determine what features evident in each phase of the MDLC can be found in each of the resulting executables.
- **Prototype of Automated Tool:** Our goal is to then develop a prototype of an automated tool that will accomplish this task when analyzing malware.
 - The results of this analysis will provide the foundation for identifying similarities in malware, specifically features common to an author or group.

Challenges

- **Obfuscation:** It is difficult to detect common features among malware when the malware is obfuscated by its author or authors.
- **Author Variations:** An author or group of authors of malware may purposefully make variations in implementation language, configuration options, and other features when writing each new piece of malware to prevent this type of analysis from proving effective.
- **Features Not Detected in Final Product:** Some key “fingerprints” found in a certain phase of the MDLC may not be retrievable in the resulting executable that will be analyzed.

Plans for the Future /
Expected Outcome

Schedule

February 7: Literature Review

February 14: Proposal Due

February 14: Understand phases of Malware Development Lifecycle and develop plan

February 21: Set up Malware Analysis Environment

February 21: Start Malware implementation

February 28: Finish 1st Malware implementation

March 6: Finish 2nd Malware implementation

March 13: Finish 2nd Malware implementation

Spring Break: Research packaging/obfuscating

Brainstorm features from malware that indicate malware similarities

Finalize components from MDLC that will be used in analysis

April 3: Midterm Report Due

April 10: Develop analysis tool

April 17: Develop analysis tool

April 24: Finish analysis tool, Review Final Report

May 1: Final Deliverables Due

Expected Outcome

- **MDLC Analysis:** We will consider the process of a software developer in each of the phases of the MDLC and determine what we can extract from looking at the end product.
 - Some additional things we will assess include the knowledge or skill level of the malware's author and links between phases of different malware that are similar due to unique choices in development and operation.
 - In addition to the information found in each phase of the life cycle, we must also consider obfuscation in our analysis, such as heavily obfuscated namespaces, source code obfuscated by multiple layers, and control flow obfuscation where a single function could be broken into multiple functions and indirect calling.
- **Expected Outcome:** Our goal is to create a prototype that automates the analysis of features found in the phases of the Malware Development Life Cycle (MDLC) as seen in the final product or malware executable.
 - From this analysis, we will attempt to identify similarities in malware.
- **Broader Impact**
 - We will develop a prototype of an automated technique to identify similar characteristics in malware based on attributes found in different phases of the Malware Development Life Cycle.
 - This will provide insight into how malware is created and common adversarial techniques which can be used to detect the authors or operators of new malware and how it is different from previous malware.

References

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